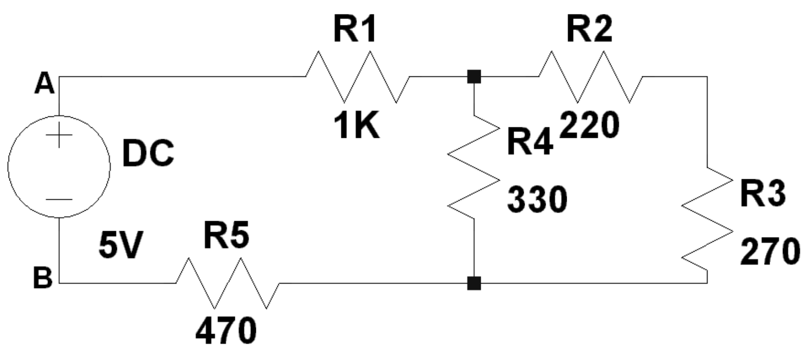
**ISLAMIC UNIVERSITY OF TECHNOLOGY (IUT)**
**ORGANISATION OF ISLAMIC COOPERATION (OIC)**
**DEPARTMENT OF ELECTRICAL AND ELECTRONIC ENGINEERING**

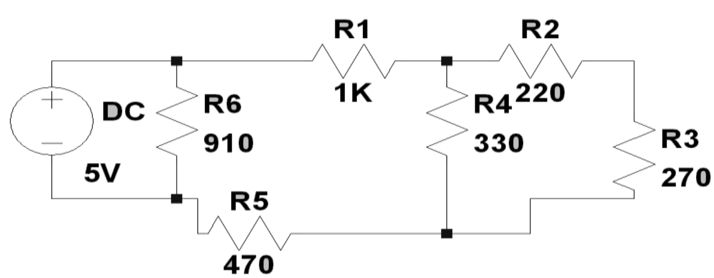
**COURSE:** EEE 4144 (PHYSICS II LAB)
**EXPERIMENT NO. 1**
**NAME OF THE EXPERIMENT:** FAMILIARIZATION WITH SIMPLE ELECTRICAL CIRCUITS.

**OBJECTIVE**:
The purpose of this experiment is to acquaint the students with the fundamentals of electrical circuits. This experiment will help students to perform future experiments. Through this experiment they will learn how to construct circuits and draw circuit diagrams.
The other objective of this experiment is to make students familiar with commonly used measuring equipment.

**LABORATORY TASKS**
**1.** Construct the following circuits and study.
**(a)**




i) Name the two sides of the DC source A & B. Does the circuit work?
ii) Measure the individual resistance and the equivalent resistance between A and B. Does it agree with the theoretical calculation?
iii) Comment on your result.
**N. B. Before measuring equivalent resistance, disconnect the DC source.**
**b)** Construct the following circuit:




i) What is the difference of this circuit with the circuit of (a)?

ii) Measure current through each branch.
iii) In which branch current is maximum?

**2.** Check whether you have taken all necessary data.

**3. Before leaving, show the completed data sheet to your teacher. You must include the data with your final report.**

**<u>Sample Data sheet:</u>**

| Name of the circuit element | Value | Current/Voltage |
|---|---|---|
| | | |
| | | |
| | | |
| | | |
| | | |

**Report:**

**i)** Answer all the questions.
**ii)** Show all calculations.
**iii)** For each setup identify which elements are connected in series and which are connected in parallel.
**iv)** What would be the problem if DC source were not disconnected before measuring the equivalent resistance by multimeter?
**v)** Discuss on your result.

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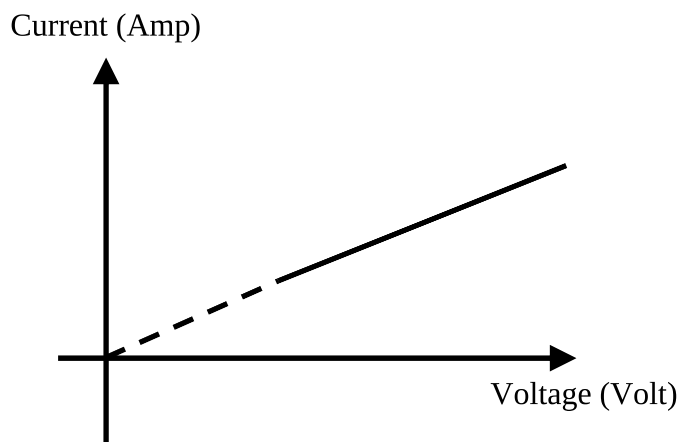
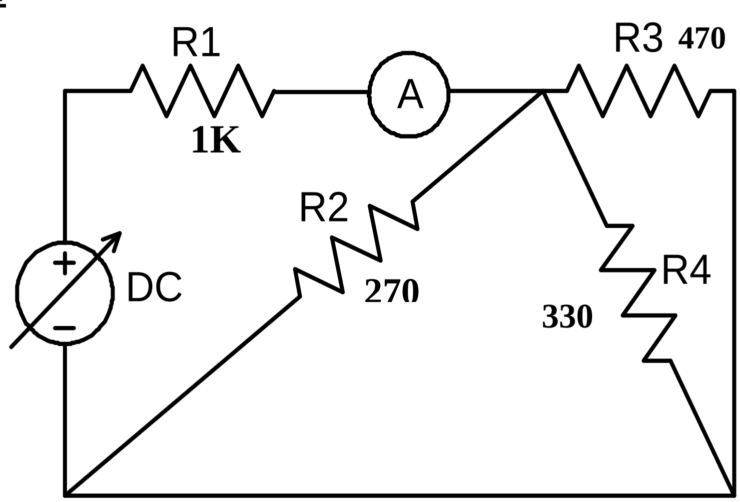
**COURSE:** EEE 4144 (PHYSICS II LAB)
**EXPERIMENT NO. 2**
**NAME OF THE EXPERIMENT:** STUDY OF OHM'S LAW

**OBJECTIVE:**
The purpose of this experiment is to introduce **Ohm's** law to the students. After completion of this experiment the students will be able to understand the importance of **Ohm's** in electrical circuits.

**Ohm's Law:** *At any constant temperature the current through any conductor is directly proportional to the voltage across* it.
If the voltage is represented by V and current by I then according to ***Ohm's law:***
$I \propto V$
or, $I = V/R$
where $1/R$ is a constant and R is the resistance of the conductor.




**Circuit Diagram:**



## Laboratory Tasks:

- Calculate maximum allowable DC voltage for the above circuit. The power rating of each resistor is 0.25Watt.
- Measure individual resistances and record their values. Construct the circuit.
- Measure current through R1 resistor and voltage across the same for different values of supply voltage up to maximum allowable voltage calculated before (increment by 2V at each step from 0V).
- Submit completed data sheet to your teacher for verification.

## Sample Data Sheet

| Supply Voltage | Voltage across R1 | Current Through R1 |
|---|---|---|
| 0 V | **0** | **0** |
| 2 V | . | . |
| 4 V | . | . |
| 6 V | . | . |
| . | . | . |
| . | . | . |
| 14 | . | . |

## Report

i. Plot *Current* vs. *Voltage* and explain the shape. The plot should be straight line and if extrapolated it will pass through origin. What does the slope of the line represent?
ii. Define *conductance* and *resistance*.
iii. Calculate resistance from the curve and compare with measured resistance value. Show calculations in your report.
iv. Discuss on your result.

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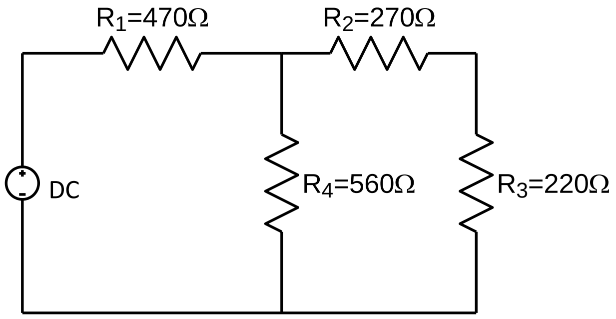
**COURSE:** EEE 4144 (PHYSICS II LAB)
**EXPERIMENT NO. 3**
**NAME OF THE EXPERIMENT:** VERIFICATION OF KVL

**OBJECTIVE:**
This experiment is intended to verify Kichhoff's Voltage Law (KVL) with the help of series-parallel circuits.

**THEORY:**
***Around any closed circuit the algebraic sum of voltage rises equals the algebraic sum of the voltage drops.***

**CIRCUIT DIAGRAM:**




**LABORATORY TASKS:**
i. Construct the above circuit.
ii. Set source voltage at 5V.
iii. Measure voltage across each resistor.
iv. Note down data in a table as shown below.

**DATA:**

| No of observation | Source voltage | Voltage across | | | |
|---|---|---|---|---|---|
| | | R1 | R2 | R3 | R4 |
| | | | | | |

**CALCULATION:**
Verify whether your data agrees with KVL.

**REPORT:**
(a) If any discrepancy is found, then explain.
(b) Can KVL be applied to open circuits?
(c) Discussion.

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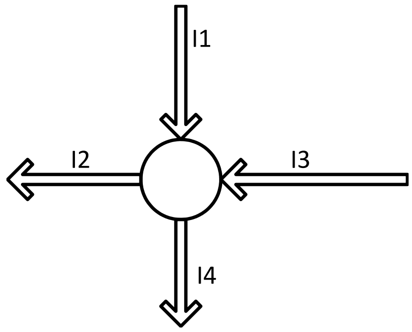
**COURSE:** EEE 4144 (PHYSICS II LAB)
**EXPERIMENT NO. 4**
**NAME OF THE EXPERIMENT:** VERIFRICATION OF KCL.

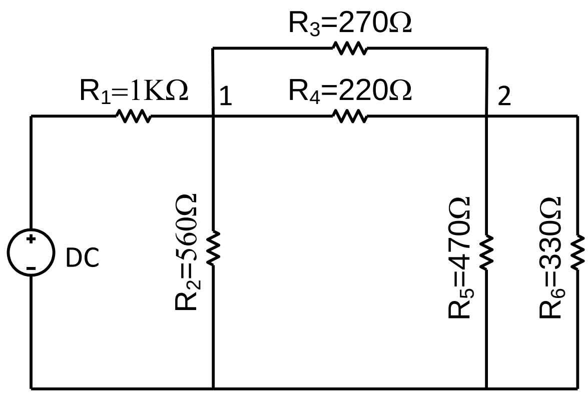
**OBJECTIVE:**
This experiment is intended to verify Kirchhoff's Current Law (KCL) with the help of series-parallel circuit.

**THEORY:**
***The algebraic sum of currents entering any node equals the sum of the currents leaving the node.***




In the figure above *I1 + I3 = I2 +I4*

**CIRCUIT DIAGRAM:**




**LABORATORY TASKS:**
i. Construct the circuit with supply voltage 10V.
ii. Measure current through each branch and note down the values.

**REPORT:**
a. Verify whether your data agree with KCL for node 1 & 2.
b. What do you understand by super node?
c. Discussion.

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**COURSE:** EEE 4144 (PHYSICS II LAB)
EXPERIMENT NO. 5
**NAME OF THE EXPERIMENT:** VERIFICATION OF THEVENIN'S THEOREM

**OBJECTIVE:**
**i)** To be familiar with the Thevenin's theorem and know its applicability.
**ii)** To verify this theorem with the help of simple network.

**Theory:**
It is often desirable in circuit analysis to study the effect of changing a particular branch element while all other branches and all other sources in the circuit remain unchanged. Thevenin's theorem is a technique to this end. It states that, 'any two terminal linear bilateral network can be replaced by an equivalent circuit consisting of a voltage source, $V_{th}$ and a series resistance, $R_{th}$.'

The value of $V_{th}$ is the open circuit voltage across the terminals and $R_{th}$ is the resistance measured between the terminals with all the sources eliminated. The voltage sources are eliminated by shorting their terminals and the current sources are eliminated by opening their terminals.

**Apparatus:**
- Four resistors (2.2KΩ, 2.2KΩ, 2.2KΩ, 3.3KΩ, 560Ω)
- One ammeter and multimeter
- Project board
- Connecting wires.

**Block diagram:**

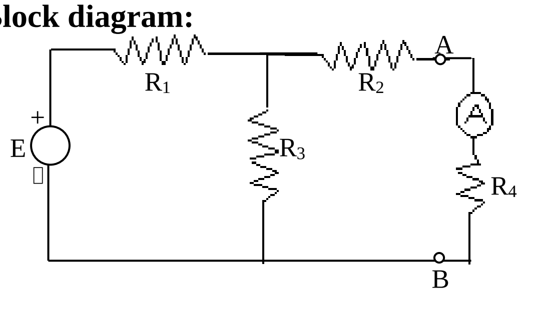


Fig. 5(a)

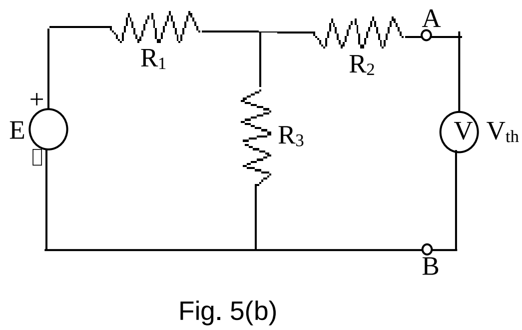


Fig. 5(b)

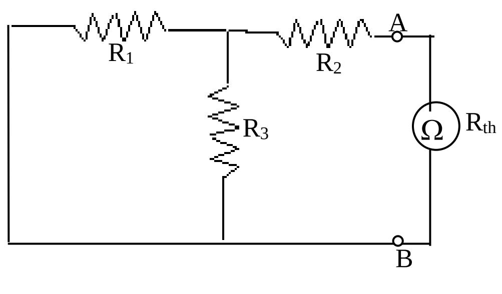


Fig. 5(c)

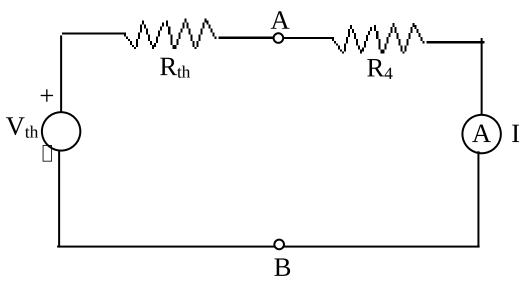


Fig. 5(d)

**Procedure:**

**i)** Implement the circuit of fig. 5(a) on the bread board taking E=10V, R1=2.2K, R2=2.2K, R3=2.2K, R4=560Ω. Take reading of current I through R4.

**ii)** Open the terminals A,B. Connect a voltmeter across the terminals as shown in fig. 5(b). Take the reading, $V_{th}$.

**iii)** Eliminate the source and short the terminals C, D and place an ohmmeter across the terminals A, B as shown in fig. 5(c). Take the reading, $R_{th}$.

**iv)** By using the voltage $V_{th}$ and resistance $R_{th}$, implement the circuit as in fig. 5(d). Take the reading of I. If the reading of I of fig. 5(a) is same that of fig. 5(d), then the Thevenin's theorem will be verified.

**v)** From the exact values of R1, R2, R3, R4 and E, calculate I theoretically. Compare this value with that of practical value.

**Data table:**

| Observation No. | I (ma) Theoretical | $V_{th}$ volts | $R_{th}$ ohm | I (ma) Practical |
|---|---|---|---|---|
| | | | | |
| | | | | |

**Report:**

**1.** Show the table to your teacher.

**2.** Comment on the obtained results and discrepancies (if any) in your discussion.

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**COURSE:** EEE 4144 (PHYSICS II LAB)
**EXPERIMENT NO. 6**
**NAME OF THE EXPERIMENT:** VERIFICATION OF SUPERPOSITION THEOREM

**OBJECTIVE:**
**i)** To be familiar with the superposition theorem and know its applicability.
**ii)** To verify this theorem which is an analytical technique of determining currents in a circuit with more than one source.

**THEORY:**
It states that, 'in any linear, bilateral network consisting of more than one source of emf, the current through or voltage across any element is equal to the algebraic sum of the currents or voltages produced independently by each source.'

While the current due to a particular source of emf is being found the other emf sources are rendered inactive and if any branch element is in series with those sources that remain intact.

**APPARATUS:**
- DC power supplies of suitable ratings
- Four resistors (6.8K, 3.3K, 2.2K, 560Ω)
- One ammeter, one multimeter
- Project board
- Connecting wires

**BLOCK DIAGRAM:**

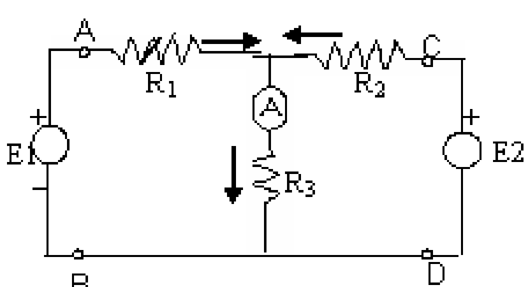

Fig. 6(a)

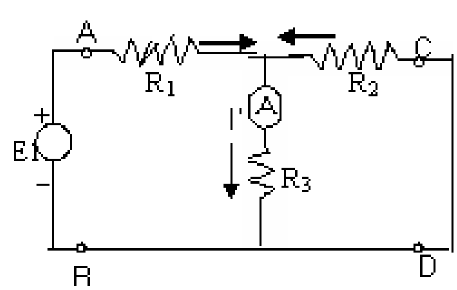

Fig. 6(b)

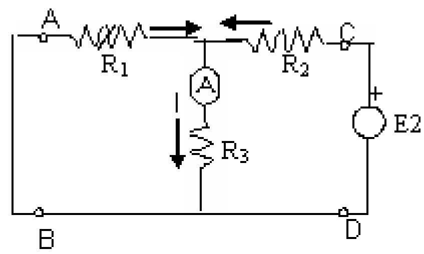


Fig. 6(c)

**PROCEDURE:**

1. Connect the circuit as shown in figure 6(a). In this circuit, keep both sources active. Apply +5 volt from E1 and -15 volts from E2 and R1 = 2.2K, R2 = 3.3K, R3 = 560Ω.
2. Measure the current I using ammeter and record it in the given table along with the current direction (upward or downward).
3. Now remove E2 source, short the terminals C, D as shown in fig. 6(b). Measure the current I´ in the branch R3 using ammeter. Record in the table.
4. Now remove E1 source, short the terminals A, B and reconnect E2 as shown in fig. 6(c). Measure the current I´´ in the branch R3 using ammeter. Record in the table.
5. Verify I = I´ + I´´ (algebraic sum) which would validate the superposition theorem for this particular circuit.
6. Repeat (2) to (5), by changing the value of R1 to 6.8K.

**DATA TABLE:**

| Observation No. | Values of R1, R2, R3 (ohms) | I (ma) | I′ (ma) | I″ (ma) | I = I′ + I″ (ma) |
|---|---|---|---|---|---|
| | | | | | |
| | | | | | |
| | | | | | |

**REPORT:**

1. Show the table.
2. Comment on the obtained results and discrepancies (if any).
3. Find theoretically the current I with the reference to fig. 6(a) applying the superposition theorem considering E1=+5V and E2=–15V and R1, R2, R3 with their recorded values.

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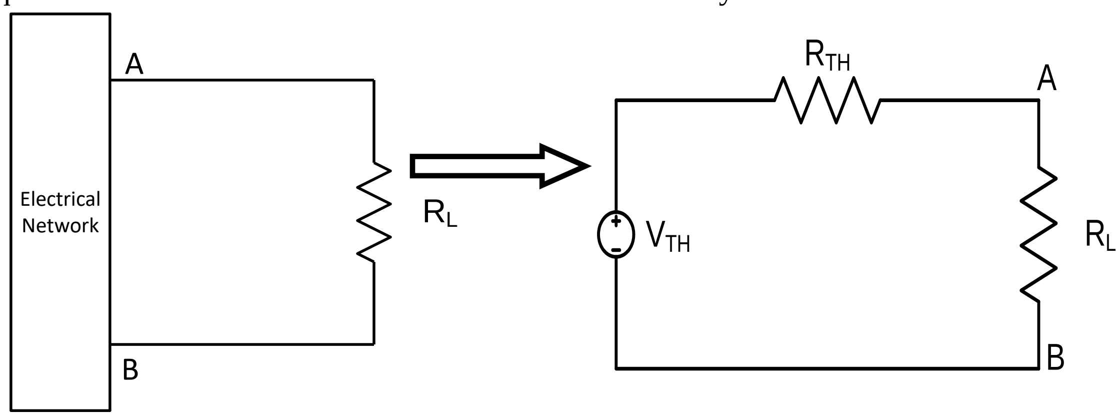
**COURSE :** EEE 4144 (PHYSICS II LAB)
**EXPERIMENT NO. 7**
**NAME OF THE EXPERIMENT:** VERIFICATION OF MAXIMUM POWER TRANSFER THEOREM

**OBJECTIVE:** The objective of this experiment is to verify maximum power transfer theorem.

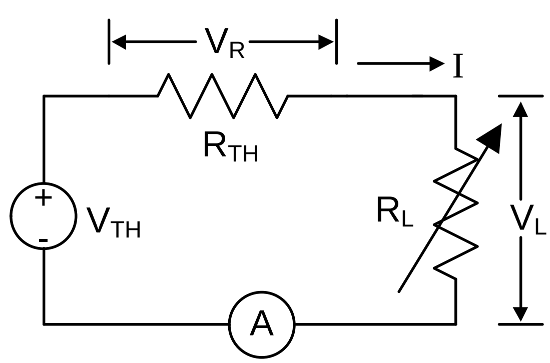
**THEORY:** "A resistive load receives maximum power when its total resistive value exactly equals the **Thevenin's** resistance of the network as seen by the load."




So, maximum power is transferred when $\boldsymbol{R_L = R_{TH}}$

**APPARATUS:**
- One 10K Potentiometer
- Two resistors (1.5K, 3.3K)
- One ammeter, one multimeter
- Project board
- Connecting wires

**CIRCUIT DIAGRAM:**



**LABORATORY TASKS:**

1. Construct the above circuit and set DC source voltage $V_{TH}$ = 15 V and $R_{TH}$ = 3.3K.
2. Vary $R_L$ from minimum to maximum value in steps and record measured $V_{TH}$, $V_R$, $V_L$, I.

**DATA :**

| No. Of Obs. | $V_{TH}$ | $V_R$ | $V_L$ | I | $P_{IN}=V_{TH}.I$ | $P_{OUT}=V_L.I$ | $R_L=V_L/I$ | $R_{TH}=V_R/I$ |
|---|---|---|---|---|---|---|---|---|
| 1 | | | | | | | | |
| 2 | | | | | | | | |
| 3 | | | | | | | | |

**REPORT:**

1. Define **efficiency of power transfer** and **voltage regulation**.
2. Calculate **efficiency of power transfer** and v**oltage regulation for** maximum power transfer condition.
3. Calculate efficiency of power transfer, voltage regulation and power loss for every observation.
4. Plot $P_{OUT}$ vs. $R_L$, %η vs. $R_L$, %voltage regulation vs. $R_L$ and loss vs. $R_L$.
5. Where maximum power transfer is used?
6. Would you suggest using maximum power transfer technique in all cases?

**FORMULAE:**

$$\text{Efficiency} = \frac{P_{OUT}}{P_{IN}} \times 100$$

$$\text{Voltage regulation} = \frac{V_{TH} - V_L}{V_L} \times 100$$

$$\text{Power Loss} = P_{IN} - P_{OUT} = IV_R$$

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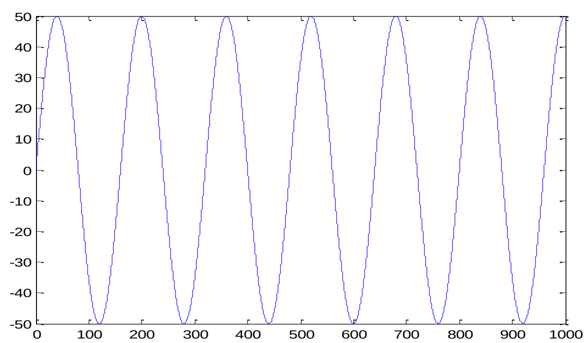
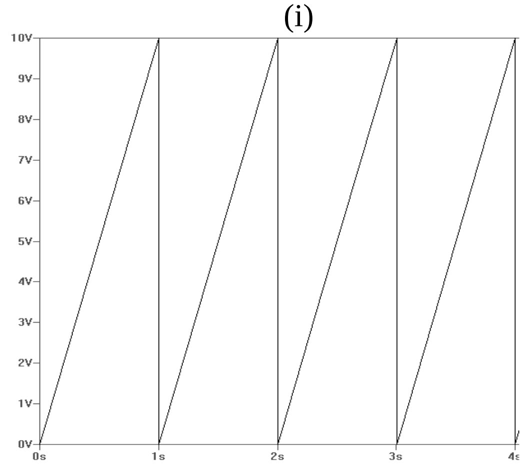
**COURSE :** EEE 4144 (PHYSICS II LAB)
**EXPERIMENT NO. 8**
**NAME OF THE EXPERIMENT:** FAMILIARIZATION WITH ALTERNATING CURRENT WAVES.

**OBJECTIVE:**
The purpose of this experiment is to make students familiar with Alternating Current (AC) waves of various shapes. Students will learn various quantities related to AC waves and also learn how to measure them by using oscilloscope.

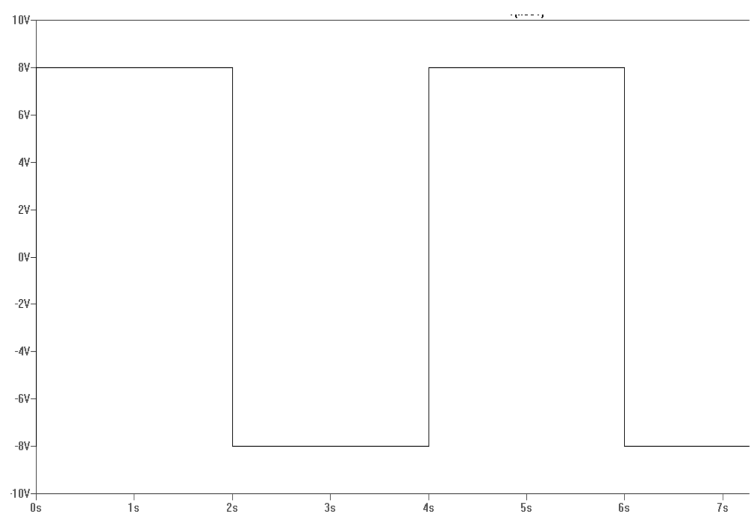
**INTODUCTION:**
Any periodic variation of current or voltage where the current (or voltage), when measured along any direction, goes positive as well as negative, is defined to be an AC quantity. The wave shapes may be of various types. The most common shapes are sinusoidal (Fig. i), triangular (Fig. ii) and square (Fig. iii).
For sinusoidal AC wave shapes the variation of current or voltage is a sine function of time.




(i)




(ii)



(iii)

**EFFECTIVE VALUES:** When we measure voltage or current by multimeter we actually get **effective** (Root Mean Square or **rms**) value. **rms** voltage is defined by

$$V = \sqrt{\frac{1}{T}\int_0^T v^2 dt}$$

Expression for current can be written in similar fashion.

**SINUSOIDAL SHAPE:**

A sinusoidal variation in time is represented by the following equation.

$$v = V_m \sin 2\pi f t$$

$$\Rightarrow v = V_m \sin (2\pi/T)t$$

In the above equation f is frequency and T is time period. Again v is instantaneous value and $V_m$ is maximum instantaneous value.

Again, a**ngular frequency** $\omega=2\pi/T$.

$$\text{So, } v = V_m \sin \omega t$$

For sinusoidal variation **rms** voltage

$$V = \sqrt{\frac{1}{T}\int_0^T V_m^2 \sin^2\left(\frac{2\pi}{T}t\right)dt} = V_m/\sqrt{2}$$

Similar is the case for current.

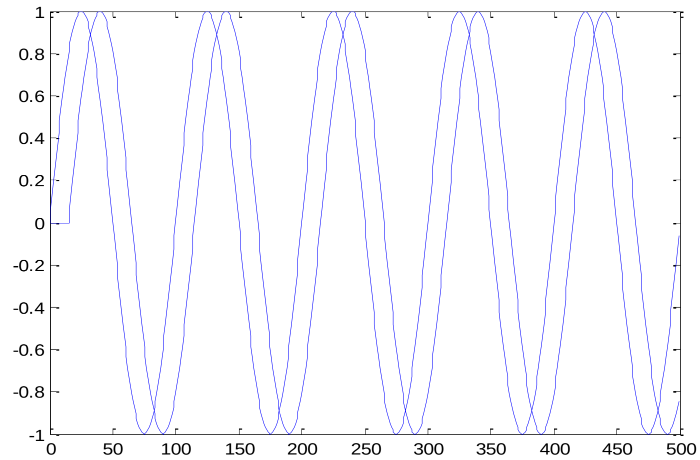
**PHASE DIFFERENCE:**

Phase difference between two ac sinusoidal waveforms is the difference in electrical angle between two identical points of two wave shapes. In the following figure voltage equations are given by

$$v_1 = V_{m1} \sin(2\pi / T)t$$

$$v_2 = V_{m2} \sin\{(2\pi / T)t - \theta\}$$

where phase difference between $v_1$ and $v_2$ is $\theta$.




**IMPEDENCE:**
Relation between voltage across and current through any component of an ac circuit is given by impedance. It is represented by Z.

$$Z\angle\theta_z = V\angle\theta_v / I\angle\theta_i$$

In AC, voltage, current and impedance all have angles associated with them.

**LABORATORY TASKS:**
1. Be familiar with oscilloscope and signal generator.
2. Activate the signal generator.
3. Generate various wave shapes of various frequencies. Draw them in your notebook. Measure time period (T) from oscilloscope. Verify whether this agree with theoretical result.
4. Measure the peak values using oscilloscope and calculate **rms** value from the mathematical equations. Measure **rms** value by multimeter. Do they agree?

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**COURSE:** EEE 4144 (PHYSICS II LAB)
**EXPERIMENT NO.:** 9
**NAME OF THE EXPERIMENT:** STUDY OF SERIES AND PARALLEL R-L-C CIRCUIT.

**OBJECTIVE:**

**i)** To be familiar with ac quantities, their phase and phase differences.

**ii)** As the ac quantities are vectors (Phasor), their additions, subtractions, and multiplications are in vector form. Hence, to be familiar with the drawing of vector diagrams of RLC series and parallel circuits.

**THEORY**

The phase of an alternating quantity is very important to locate it properly with respect to a reference. Phase is the fractional part of a period through which time or the associated time angle ($\theta = \omega t$) has advanced from an arbitrary reference.

Fig. 9(a) shows phase relation between current through and voltage across resistive, purely inductive and purely capacitive elements.

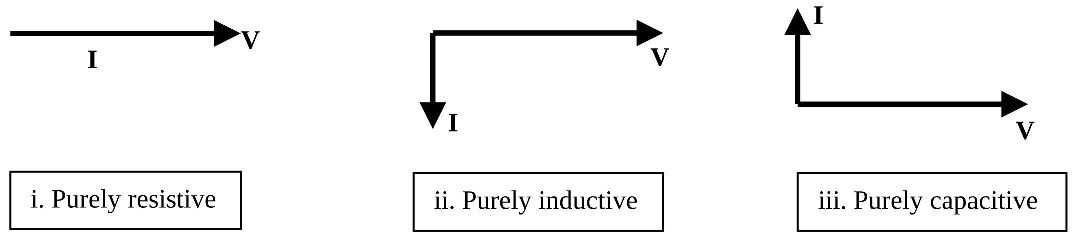


**Fig. 9(a): Vector Diagram**

In RLC series circuit, if a sinusoidal voltage is applied, then, according to KVL, the applied voltage must be equal to the vector addition of $V_R$, $V_L$, $V_C$.

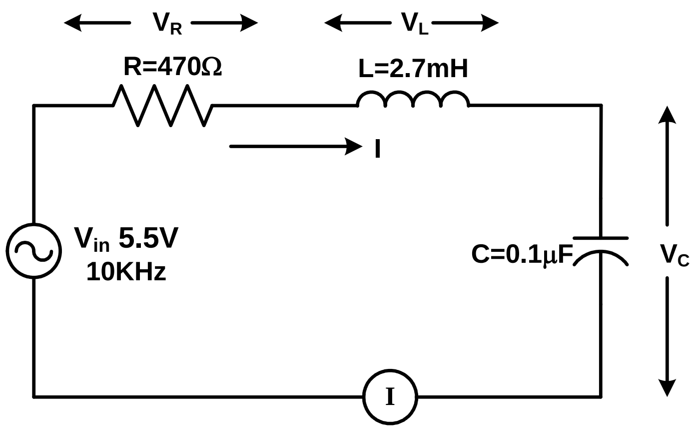


**Fig: 9(b)**

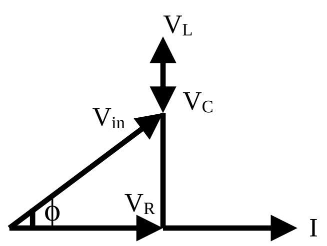


**Fig: 9(c)**

$V_{in} = V_R + V_L + V_C$

⇨ $V_{in} = IR + jIX_L - jIX_C$

$= I\{R + j(X_L - X_C)\}$

⇨ $V_{in}/I = Z = \sqrt{\{R^2 + (X_L - X_C)^2\}} \angle \tan^{-1}(X_L - X_C)/R$

Here, V and I indicate the rms values, Z is the impedance.

In a series circuit, current through the circuit is constant. Hence, to draw vector diagram of a series circuit, it is convenient to take current as reference. Conversely for parallel circuit voltage across parallel elements is taken as reference.

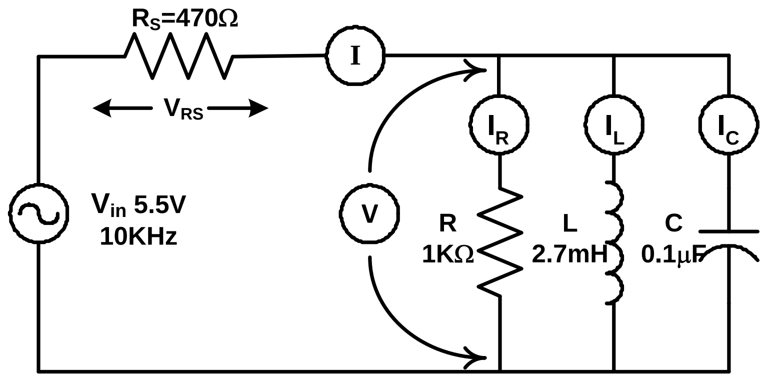


Fig: 9(d)

**APPARATUS:**

- Signal generator
- Capacitor 0.1µF
- Resistors (470Ω, 1KΩ)
- Inductor 2.7mH
- Multimeter
- AC ammeter (20mA or 10mA)
- Bread board
- Oscilloscope
- Connecting wires etc.

**PROCEDURE**

**i.** Implement the circuit of Fig. 9(b) and then set the amplitude of the AC signal to maximum ($V_{in}$ should be around 5V RMS). Take the readings of I, $V_{in}$, $V_R$, $V_L$, $V_C$ using multimeter. Observe the voltage wave shapes in oscilloscope. Measure frequency and amplitudes of the voltages. Compare the voltage readings from multimeter and oscilloscope.

**ii.** Then draw a vector diagram showing all voltages and current with the help of the method shown in Fig. 9(c).

**iii.** Similarly, connect the parallel circuit shown in Fig. 9(d) and then set the amplitude of the AC signal to maximum ($V_{in}$ should be around 5V RMS). Take the readings of $V_{in}$, $V_{RS}$, V, I, $I_R$, $I_L$, $I_C$ using multimeter. Observe the voltage wave shapes in oscilloscope. Measure frequency and amplitudes of the voltages. Compare the voltage readings from multimeter and oscilloscope.

**iv.** Then draw a vector diagram showing all voltages and currents following the hint given.

**DATA TABLE:**

(a) For series circuit:

| $V_{in}$ (Volt) | I (mA) | $V_R$ (Volt) | $V_L$ (Volt) | $V_C$ (Volt) |
|---|---|---|---|---|
| | | | | |

(b) For parallel circuit:

| $V_{in}$ (Volt) | $V_{RS}$ (Volt) | V (Volt) | I (mA) | $I_R$ (mA) | $I_L$ (mA) | $I_C$ (mA) |
|---|---|---|---|---|---|---|
| | | | | | | |

**REPORT:**

1. Show the tables and vector diagrams.
2. Discuss on the obtained results and discrepancies (if any).